

Meng Gao

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EDUCATION

Tsinghua University *Master of Engineering in Artificial Intelligence* Aug. 2024 – Jun. 2027

- **GPA:** 3.9/4.0 (Top 10%)
- **Advisor:** [Yujiu Yang](#)
- **Research interests:** reinforcement learning, efficiency, and multimodal interaction
- **Selected coursework:** Artificial Neural Network (4.0), Introduction to Statistical Learning Theory (4.0), Artificial Intelligence Programming (4.0), Selected Topics on Intelligent Information Processing (4.0)

Fudan University *Bachelor of Science in Mathematics (Talent Program)* Aug. 2020 – Jun. 2024

- **Advisor:** [Yingzhou Li](#)
- **Research interests:** Fast algorithms and quantum computing
- **Selected coursework:** Functional Analysis (H) (A), Ordinary Differential Equations (A), Computational Methods / Numerical Analysis (A-), Python Programming (A), Deep Learning (A)

INTERNSHIP

Qwen-Omni Team, Alibaba *Research Intern* (Mentors: [Yunfei Chu](#), [Jin Xu](#)) Mar. 2026 – Now

- Researched long-form audio understanding and reasoning, including benchmark design, QA construction, and evaluation of large audio-language models.
- Investigated methods to improve temporal grounding and counting abilities in omni models.
- Designed a training paradigm with per-turn loss scaling for streaming proactive-response tasks, and built an audio proactive benchmark and the synthetic data pipeline.

Microsoft Research Asia (MSRA) *Research Intern* (Mentors: [Lei Ji](#), [Yeyun Gong](#)) Jun. 2025 – Feb. 2026

- Researched efficient multimodal retrieval with a focus on inference acceleration.
- Developed token and layer pruning methods to reduce redundant computation.
- Designed adaptive early-exit mechanisms based on input difficulty.

PUBLICATIONS

Skim and Skip: Hierarchical Adaptive Inference for Efficient Multimodal Retrieval. [\[Paper\]](#) [\[Code\]](#)

Meng Gao*, Yizhen Zhang*, Yang Ding*, Ziqi Dai, Shuoshuo Zhang, Junjie Wang, Taiqiang Wu, Chufan Shi, Lei Ji, Jian Jiao, Linfeng Zhang, Yeyun Gong, Yujiu Yang **EMNLP 2026 (Main Conference)**

- Proposed Skim and Skip (SAS), a hierarchical adaptive inference framework for efficient multimodal retrieval.
- Designed token-level evidence selection and depth-adaptive early exit to reduce redundant computation.
- Achieved up to $1.64\times$ speedup and $2.97\times$ FLOPs reduction while retaining 99% retrieval performance.

AnyCap: Omni-Modal Captioning with Instruction Alignment. [\[Paper\]](#) [\[Code\]](#)

Yiming Ren*, Zhiqiang Lin*, Yu Li*, Meng Gao*, Junjie Wang, Zicheng Lin, Jifeng Dai, Yujiu Yang, Wenhai Wang, Ruihang Chu **EMNLP 2026 (Main Conference)**

- Developed AnyCap, a plug-and-play framework for instruction-aligned captioning.
- Built the audio data pipeline and contributed to AnyCapEval for fine-grained content-style evaluation.
- Improved caption quality and instruction following across both open-source and API-based models, with gains validated on public benchmarks.

SELECTED PREPRINTS & MANUSCRIPTS

ProactiveAudioBench: Evaluating Condition-Driven Responses in Audio Streams.

Meng Gao, Yunfei Chu, Yifan Yang, Hui Wang, Wenxiang Guo, Qi Chen, Zihan Liu, Ling Wang, Yinsong Yan, Wen Huang, Xize Cheng, Yuxuan Wang, Jin Xu, Yujiu Yang

- Built ProactiveAudioBench for proactive listening with Detection, Tracking, Counting, and Matching tasks.
- Evaluated models under simulated and native streaming, revealing distinct timing error patterns and strong challenges in Counting and Matching.

Where You Put It Matters: Characterizing Positional Sensitivity in Large Audio Language Models.

Hui Wang*, Meng Gao*, Yunfei Chu*, Jin Xu, Yong Qin

- Characterized positional sensitivity in multi-audio reasoning with controlled permutations.
- Showed task- and scale-dependent position effects across multiple audio-language models.

MLP: Modality-Aware Layer Pruning for Efficient Multimodal Retrieval

Minji Seo, **Meng Gao**, Zhiwei Yu, Fanhong Li, Jaesik Park, Yujiu Yang, Lei Ji

- Proposed MLP, a modality-aware layer pruning framework for efficient multimodal retrieval.
- Designed structure-guided layer selection and hardness-scaled self-distillation for pruned retrieval models.
- Reduced computation by 22% while retaining over 96% of retrieval performance.

ADDITIONAL PREPRINTS & MANUSCRIPTS

AudioSpan: Spanning the Duration and Depth of Audio Comprehension.

Wen Huang*, Yunfei Chu*, **Meng Gao**, Haolin He, Jin Xu

Omni Demand Understanding: A Benchmark for Contextual User-Intent Inference in Multimodal Interaction.

Qi Chen, Yunfei Chu, Haolin He, Yifan Yang, Zihan Liu, Yuxuan Wang, Ziyang Ma, Ruiyang Xu, **Meng Gao**, Yinsong Yan, Ling Wang, Hui Wang, Wen Huang, Yiheng Chen, Guanrou Yang, et al.

TickTock: Teaching Video LLMs to Tell Time at Frame-level Precision.

Ling Wang, Yinsong Yan, Yuxuan Wang, Yifan Yang, Junming Lin, Ruiyang Xu, Yunfei Chu, **Meng Gao**, Qi Chen, Minghao Han, Xize Cheng, Schmitt.Z, Qize Yang, Jin Xu, Jibin Wu

LAMAR-Bench: Diagnosing Long-Form and Multi-Audio Understanding and Reasoning.

Hui Wang, Yunfei Chu, Xize Cheng, Yu Xi, **Meng Gao**, Qi Chen, Yifan Yang, Wenxiang Guo, Yong Qin, Jin Xu

Omni-InteractScope: Benchmarking Proactive and Agentic Interaction in Continuous Omnimodal Streams.

Yinsong Yan, Ling Wang, Yuxuan Wang, Yifan Yang, Ruiyang Xu, Junming Lin, Qi Chen, Yunfei Chu, Qize Yang, **Meng Gao**, Xize Cheng, Minghao Han, Jin Xu, Jibin Wu

TECHNICAL REPORTS

Qwen3.8-Omni: Towards Native Omni-Modal Agents.

Qwen Team | [arXiv:2609.25611, 2026](#) | [Project Page](#) | [Preprint](#)

UNDERGRADUATE THESIS

Hash-Table Acceleration of 2RDM Computation in OptOrbFCI Algorithm

- Designed a sorted-hash-table algorithm with binary-search lookup to accelerate two-body reduced density matrix (2RDM) computation in full configuration interaction (OptOrbFCI), a quantum-chemistry solver.
- Analyzed the lookup scheme’s complexity and benchmarked it against the open-addressing hash baseline.
- Achieved a 45% average reduction in computation time with more stable runtime.

TEACHING ASSISTANT EXPERIENCE

Teaching Assistant, Mathematical Analysis II, Fudan University (Spring 2023)

AWARDS & HONORS

Stars of Tomorrow, Microsoft Research Asia Internship Program, 2026
Outstanding Undergraduate Graduate, Fudan University, 2024
Fudan University Merit Scholarship (Second & Third Class) and Academic Excellence Scholarship, 2021–2023
Merit Student & Outstanding Student Leader, Fudan University, 2020–2022
First Prize – Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), 2022
Second Prize (2021) & Third Prize (2022) – National College Student Mathematics Competition (Class A)

SKILLS & INTERESTS

Programming Languages: Python, LaTeX, MATLAB, C++, R
Languages: Chinese (native), English (fluent)
Interests: Singing, Choir, Chinese Classical Dance (Level 10), Swimming